Banksy: Evidence-Gated Repair of Garbled Text in AI-Generated Videos

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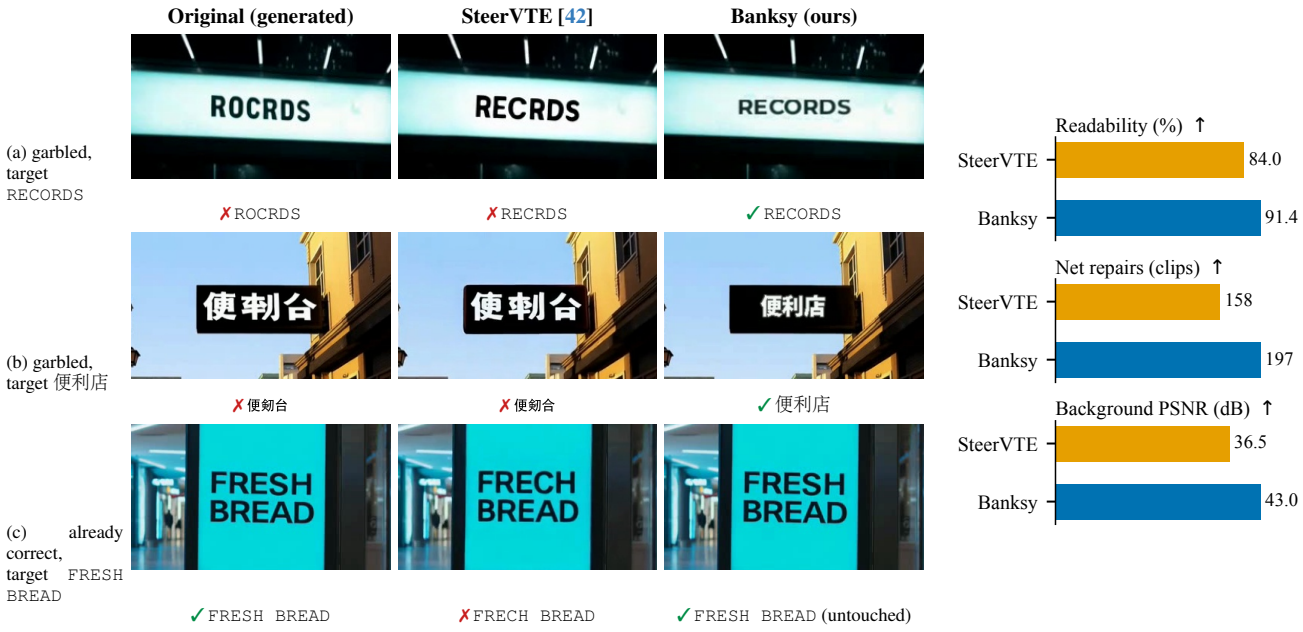


Figure 1. **Repairing garbled text in generated videos.** A generative video text editor rewrites the text region and often produces plausible but wrong characters (a, b) or breaks text that was already correct (c). Banksy edits only when evidence supports it, writes the target string deterministically, and leaves everything outside the text untouched. Below each crop: the transcription of frame 24 by the evaluation reader (✓ correct, ✗ wrong). Right: headline numbers on our 526-clip benchmark (Sec. 5.1). Examples (a, b) were chosen among clips where SteerVTE fails and Banksy succeeds; (c) is the only clip among the 46 that Banksy leaves untouched which SteerVTE breaks.

Abstract

Text-to-video generators still render garbled text on signs, screens and posters. We study repairing *such text: given a generated clip, a target string and a hint box, make the text read correctly while leaving everything else as it was. Generative video text editors produce natural-looking glyphs, but they frequently keep or invent wrong characters, rewrite text that was already correct, and re-render the background. We present Banksy, a training-free, correctness-first repair pipeline built around three evidence gates. A* preservation gate *leaves a clip untouched when multi-frame OCR already reads the target; a* localization gate *lays the new text out on the old glyph instance when the evidence is strong and falls back to the tracked box otherwise; and an* appearance gate *enables a non-parametric source appearance transport (SAT) that carries glow and material from the old glyphs to the new ones only where it measurably helps. The target string itself is rendered deterministically, so its spelling cannot drift, and only pixels inside the text region change. On a benchmark of 526 generated clips spanning seven carrier materials in English and Chinese, Banksy reaches 91.4% readability against 84.0% for the state-of-the-art generative editor SteerVTE (197 vs. 158 net repairs) and keeps the background intact (43.0 vs. 36.5 dB PSNR outside the text region). In a blind study, raters preferred SAT for material fidelity in 87% of non-tied judgements. We also report, with pre-registered criteria where applicable, what did*

not work—a learned 1.3B refiner, glyph-only erasure and a carrier-fit check—and we characterise the remaining appearance gap to generative editing.

1. Introduction

Text-to-video models such as Wan [31], CogVideoX [41] and VACE [16] synthesise convincing street scenes, shop fronts and screens, yet the text inside them is often garbled: a record shop reads ROCRDS, a convenience store reads 便刻台 (Fig. 1). In our benchmark of generated clips, only 54.0% of the target strings are rendered correctly. For advertising, signage, education or any use in which the text carries meaning, a single wrong character is a hard failure, and re-generating the clip until the text happens to come out right is expensive and changes the shot. We therefore study *repair*: keep the generated clip, and make its text read correctly.

The natural tool is a generative video text editor [42], which regenerates the text region conditioned on rendered glyphs and on the style of the source. Such editors produce glyphs that blend into the scene, but on generated videos we observe three systematic problems. They write characters that look plausible but are wrong—FRESY BREAD, 小心坍滑, 洹影院—and on our benchmark the strongest of them, SteerVTE [42], still leaves 16% of the clips unreadable. They rewrite text that was already correct, turning FRESH BREAD into FRECH BREAD (Fig. 1c). And because the whole clip passes through the generator, the background changes as well: the PSNR outside the text region drops to 36.5 dB.

We argue that repair should be *correctness-first* and *minimal*, with a strict order of priorities: do not change what is already right; when a change is needed, change only the text; and recover the material of the old text only when there is evidence that this helps. Banksy implements this order with three evidence gates (Fig. 2). A preservation gate returns the clip unchanged when OCR on many frames already reads the target. A localization gate decides whether the new text can be laid out on the old glyph instance—its size, colour and baseline—or must fall back to filling the tracked box. An appearance gate decides whether to apply source appearance transport (SAT), a non-parametric operator that carries glow and material from the old glyphs to the new ones. Between the gates, the target string is rendered deterministically from a matched font and warped with carrier homographies, so its spelling cannot drift, and only pixels inside the text region are composited. The pipeline requires no training.

On 526 generated clips covering seven carrier materials in English and Chinese, Banksy raises readability from 54.0% to 91.4% (SteerVTE: 84.0%), with 197 net repairs against 158, and keeps the background intact (43.0 dB). We are equally explicit about what it does not do. Its glyphs are rendered, not generated: where both methods write the target correctly, SteerVTE's text is closer to the original appearance in 70% of the clips, and Banksy breaks 14 originally correct clips where SteerVTE breaks 6. We trace these gaps to their causes (Sec. 6.2) and report the alternatives we tried and rejected.

Our contributions are:

- **A repair task, benchmark and protocol.** 526 generated clips with hint boxes and targets, seven materials, two languages; metrics that separate repairs from breakages (net repair), background fidelity and collateral change, temporal stability, and blind human judgements. Development clips are disclosed and excluded in a separate report, and the selective-editing study was pre-registered.
- **Banksy**, a training-free, evidence-gated repair pipeline: preservation and localization gates, deterministic glyph rendering anchored to the tracked carrier, and SAT behind an appearance gate.
- **An extensive analysis including negative results**: a learned 1.3B refiner trained on up to one million pairs does not improve appearance and costs readability and temporal stability; glyph-only erasure and a carrier-fit check fail their pre-registered criteria; and a failure taxonomy that locates the remaining appearance gap.

2. Related work

Scene text editing in images. Early editors decompose a text image into background, foreground style and content and re-synthesise it with GANs [17, 27, 28, 36]; STEFANN [28] generates missing characters in the font of an observed one. Diffusion models brought text rendering and editing to arbitrary images through glyph conditioning [4, 15, 21, 32–34, 40, 43, 47]. These methods produce natural glyphs, but spelling remains a sampling outcome rather than a guarantee, and in video, frame-wise application produces flicker. A related line renders text first and harmonises it afterwards [37, 44]; our learned refiner (Sec. 5.6) belongs to this family.

Text in generated and edited videos. Text-to-video models [16, 31, 41] render legible text only occasionally: dedicated benchmarks find most open and commercial models unable to produce legible, temporally consistent text [11, 13, 45, 49], and image and video tokenizers already lose small text in reconstruction [35]. Generation-time remedies condition on glyphs [19] or fine-tune on synthetic text-rich clips [22]; they lower the rate of garbled text but do not repair a given clip. For editing, SteerVTE [42] injects line- and character-level glyph features and a VLM style embedding into a frozen VACE backbone and trains on a million synthetic and real triplets. VidScribe [46] evaluates text

generation and localised editing in videos, including commercial models, and finds editing the weakest regime, with edits often corrupting text that should not change; ViTeX-Bench [5] evaluates video scene-text editing, including edit locality, and releases a 14B reference editor. All of these regenerate the text region. We instead repair generated text with a training-free pipeline that leaves already-correct text untouched, writes the target deterministically and changes only the text region, and we compare against SteerVTE as the strongest generative editor.

Text effects transfer. Statistics-based and example-based methods transfer the appearance of stylised text to new glyphs [25, 38, 39], using distance to the glyph boundary as the key coordinate. SAT follows this idea but fits a light-transport operator on the old glyphs of the same video and gates its use on measured reconstruction gain.

Video inpainting and consistent editing. Removing the old text is a video inpainting problem [3, 48]; we inpaint once on a reference frame with LaMa [30] and propagate the result along the tracked carrier, which avoids per-frame hallucination. Flow- and correspondence-guided attention improves the temporal consistency of diffusion-based video editing [6, 10]; we test a carrier-anchored attention bias of this kind on our learned refiner (Sec. 5.6).

Evaluating text. OCR systems [1, 7, 29] and, increasingly, vision-language models [2] are used to score generated text. We use a VLM reader for evaluation and a separate OCR inside the pipeline, report repairs and breakages separately, and complement automatic scores with blind human studies.

3. Task, benchmark and metrics

Task. The input is a generated clip V, a target string s and a hint box b that marks where the text is. In our benchmark the target is the quoted sign text of the generation prompt, and the hint box is produced automatically by OCR localisation and tracking of the generated text; the same box is given to every method, and SteerVTE uses it as its editing mask. The output is a clip $\hat{V}$ of the same size and length in which the text inside b reads s and nothing else has changed. Unlike scene text editing, the source text is usually *not* a valid word—it is a near miss produced by the generator—and in about half of the clips it is already correct, so "do nothing" is often the right answer. We call the task *repair* to stress both aspects.

Benchmark. We generated short clips of street scenes, shop fronts and screens with the text-to-video mode of VACE-14B [16], from 120 prompts that ask for a specific sign text on one of seven carrier materials, with ten seeds each (832×464 pixels, 49 frames at 15 fps). Of 1,178 usable clips, an OCR screen confirmed the requested text in only 24%. From the rest, an EasyOCR-based screen selected 526 clips as garbled; this screen is stricter than our evaluation reader, which judges 54.0% of them correct, so the benchmark contains both repair and do-nothing cases (exact selection rule to be documented). The 526 clips come from 117 prompts with up to ten seeds each, 40 target strings (20 English, 20 Chinese, from one to four words or characters, *e.g.* `BUS STOP`, `SALE 50%`, 小心地滑), and seven carrier materials: neon (93 clips), signboard (86), paper (96), glass (76), screen (65), metal (56) and wood (54); 295 clips are English and 231 Chinese. Every clip has a hint box per frame. The reader (below) judges 284 originals (54.0%) correct and 242 garbled. Forty clips were looked at while developing the selective-editing components; they are listed in the supplement and excluded in a separate report.

3.1. Metrics

Readability. A vision-language reader, Qwen2.5-VL-7B [2], transcribes the text inside the hint box (15% padding) on 5 frames spread over the clip. A clip counts as *correct* if at least 4 of the 5 transcriptions match the target exactly after removing whitespace and line breaks, lower-casing, and folding traditional to simplified Chinese; readability is the share of correct clips, and NED similarity the mean normalised edit similarity over the 5 frames. The reader is independent of Banksy, whose gates use a different OCR system. All outputs that are compared are read in the same job and software environment.[1]

Repairs and breakages. Readability alone hides two opposite effects, so we also count *fixed* clips (garbled → correct), *broken* clips (correct → wrong) and their difference, the *net* repairs.

Locality. *BG PSNR* is computed between output and input on all pixels outside the hint box dilated by half its height, over 49 frames. *Collateral change* is the mean absolute grey-level change, on the reference frame, inside the text neighbourhood (box grown by half its height) but outside both the old glyphs and the new glyphs (each dilated by 0.15 box heights); it isolates damage that is not part of writing the new text, and it requires the method's glyph mask, so we report it for Banksy only.

Appearance. The original glyphs are wrong but their material is the reference. On three frames, we estimate an

[1] We learned this the hard way: one version of the reader produced garbled transcriptions for about 8.5% of the inputs on one cluster and none on another; comparing outputs read in different environments inflated an early result (supplement).

ink mask inside the box for every video and compare, in CIELAB, the ink colour, the ink–ground contrast, the halo around the strokes (normalised by contrast), the edge softness and the shading inside the strokes with those of the original. Each feature is divided by its median over all method–clip pairs of the evaluation run and the five are averaged; lower is closer. The normalisation makes values comparable only within one run, which is why every table compares variants from a single run.

Temporal stability. The carrier's frame-to-frame motion is estimated once per clip on the original (SIFT, RANSAC homography) and applied to every output, so camera motion cancels. *Flicker* is the mean absolute motion-compensated grey-level change inside the box, *jitter* the same for gradient magnitudes (stroke position), and *boundary jump* the change across the frame pairs at which a 17-frame chunked model switches chunks, relative to all other pairs (1 means no seam; used for the learned refiner in Sec. 5.6).

Human judgements. Blind pairwise studies show the original next to two versions in random left/right order, with the target known to the raters; questions are asked separately (*e.g.* material fidelity, artifacts, flicker), and ties are allowed. Study designs, sampling and rater quality control are in the supplement.

4. Banksy

Banksy takes a generated clip $V = \{I_t\}_{t=1}^{T}$, a target string s and a hint box b (Sec. 3) and returns a clip $\hat{V}$ that differs from V only inside the text region. Figure 2 summarises the pipeline; all components are classical or pre-trained and nothing is trained for the task. We describe the steps in order and give thresholds and implementation details in the supplement.

4.1. Reading the text and tracking its carrier

We run an off-the-shelf OCR system (EasyOCR [14], CRAFT detection [1] and CRNN recognition [29]) on every frame and keep the boxes that overlap b. Boxes are grouped into text rows, and the frame whose rows best match s (normalised edit similarity) becomes the reference frame $t^\star$. The text is printed on, or emitted by, a physical *carrier*—a sign, a poster, a screen—so we track the carrier rather than the glyphs: SIFT keypoints [23] are matched between $I_{t^\star}$ and every I_t inside a window one text-box size larger on each side of the reference quad, and a RANSAC [9] homography H_t is estimated (ratio test 0.75, reprojection threshold 2.5 px, at least 10 inliers). When a frame fails, H_t falls back to the quad-to-quad homography between the OCR boxes. Everything that is placed on the carrier later—erased background, glyphs, masks—is placed once in the reference frame and carried to frame t by H_t, which removes per-frame jitter of the OCR boxes from the output.

4.2. Preservation gate

Generated text is often already correct, and any edit can only make such a clip worse. The preservation gate returns $\hat{V} = V$ when the OCR evidence across frames shows that the clip already reads s: the complete target is read in at least 85% of the frames, wrong readings occur in at most 5%, the reading lies inside the hint box, at most 10% of the frames contain extra boxes, and the reading of *all* boxes inside the hint box, concatenated in reading order, equals s in at least 90% of the correct frames. The last condition prevents skipping when the target appears next to other, garbled text (*e.g.*, a correct word above a garbled subtitle). The gate uses only the OCR of the pipeline, never the evaluation reader (Sec. 3.1), and when the evidence is insufficient the clip is edited: the gate trades recall for safety. In the final run it skips 46 of the 526 clips (9%), all of which the independent reader confirms as correct.

4.3. Localization gate and layout

If the old glyphs can be localised reliably, the new text should inherit their size, colour and baseline. We estimate a glyph *instance* in the reference frame: the ink mask (colour clustering inside the rows), the ink and ground colours, the letter height, the number of letters and the letter band. The localization gate accepts the instance only if it contains at least three letters (two for Chinese), the local contrast between ink and inter-letter ground exceeds 40 in CIELAB, the letter height is between 0.3 and 1.0 of the OCR box height, and the OCR rows did not require a fallback grouping. Otherwise the clip falls back to a *box layout* that fills the tracked box with legible text. In the final run 38% of the clips take the box layout.

With an accepted instance, the target is laid out on the letter band: the cap height equals the measured letter height, the string keeps its natural aspect ratio in the selected font, and it is centred on the old text; a target wider than the old line may extend the line by up to $1.35\times$ before its height is reduced. The band is oriented like the OCR quad, which for most detectors is axis-aligned—a limitation we return to in Sec. 6.2. Stacked text keeps its rows: the target is split into lines following the word counts of the detected rows, and each line is laid out on its own row instance.

4.4. Erasure and deterministic rendering

Erasure. The old glyphs are removed once, in the reference frame: the ink mask (closed, and extended by the bright pixels of glowing text) is dilated by $\min(0.3h, \max(10, 0.18h))$ pixels for text height h, flat grounds are pre-filled with their median colour, and the hole is inpainted with LaMa [30]. The erased background is then

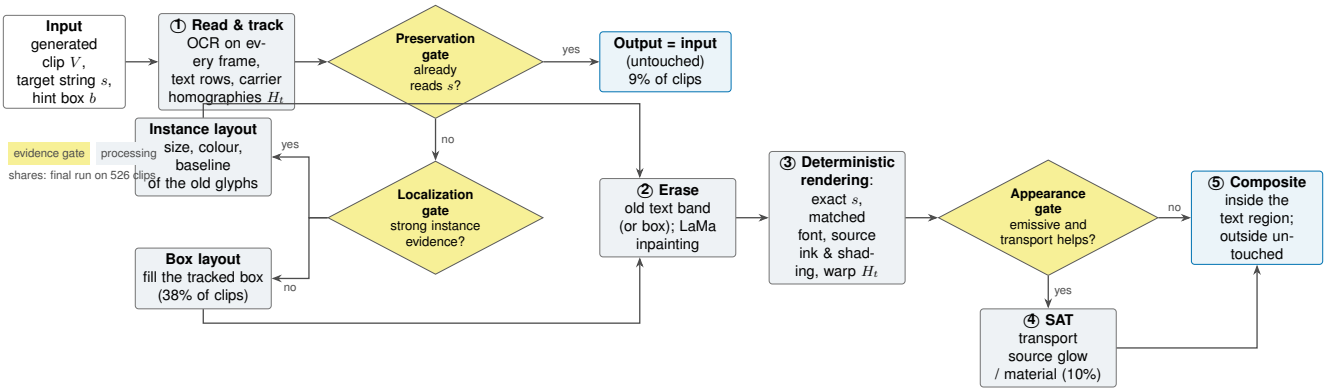


Figure 2. **Banksy overview.** Three evidence gates decide how much to change. The preservation gate returns the input unchanged when multi-frame OCR already reads the target; the localization gate chooses between laying the new text out on the old glyph instance and filling the tracked box; the appearance gate enables source appearance transport (SAT) only on emissive carriers where it measurably reduces the reconstruction error. The target string is always rendered deterministically, so its spelling cannot drift, and only pixels inside the text region are composited.

carried to every frame by H_t, with a per-frame, per-channel gain fitted on a ring around the hole and smoothed over five frames, so that exposure changes follow without per-frame inpainting flicker.

Font matching. The new glyphs are rendered, not generated. The font is chosen from a library of 33 Latin and 8 Chinese families (variable fonts expanded over their weights) by comparing stroke statistics of the warped old ink mask with those of each candidate rendered in the same layout; script and display faces are admitted only when they are clearly better (distance below 0.6 of the best plain face). Because the target string is drawn by a font rasteriser, its spelling is exact by construction.

Photometry. The glyph colour follows the per-frame ink colour of the old text (running median over seven frames) modulated by a low-frequency shading field estimated from the old strokes in canonical text coordinates, and glowing text receives a soft halo. The canvas is warped to every frame with H_t and composited over the erased background.

4.5. Appearance gate and source appearance transport

Rendering gives the right letters, but on emissive carriers—neon, lightboxes, screens—the old glyphs carry glow, bloom and inner structure that a flat fill cannot reproduce. SAT transports this appearance from the old glyphs to the new ones without regenerating the region. Let B be the erased background, I the reference frame and $d(x)$ the signed distance of pixel x to the boundary of the old glyphs, in units of the letter height. On the old text we fit, per distance bin k and per channel, a multiplicative-plus-additive light-transport operator

$$I(x) \approx m_k\, B(x) + a_k, \qquad k = k(d(x)), \tag{1}$$

over the support $d \le 0.6$, and apply it to the new glyphs with their own distance field, $\hat{I}(x) = m_{k(\hat{d}(x))} B(x) + a_{k(\hat{d}(x))}$. On the old text, this family reconstructs the source better than an additive-only operator, a distance-indexed replacement or a flat fill (cross-validated CIELAB error 17.2 vs. 25.9, 27.6 and 29.7; best on 45 of 52 clips; supplement), which is why we keep both terms.

SAT is applied only behind an *appearance gate* with two conditions: the text must be emissive (ink brighter than its ground, or a detected glow on the carrier), and the operator must generalise across the text—fitted on one half and tested on the other, in both directions, its error must stay below 0.75 of the flat-fill error. In the final run, SAT is enabled on 51 clips (10%); it is never applied to box-layout clips, whose instance evidence is weak by definition.

4.6. Compositing

The output differs from the input only where the erasure mask or the new glyphs (with halo) are non-zero; elsewhere the input pixels are copied, so the only residual outside the text region is the re-encoding of the video.

5. Experiments

Protocol. All numbers in this section come from one front-end pass over the 526 benchmark clips in which every variant of Banksy shares the OCR, tracking and instance estimation, so variants are paired clip by clip; all methods (including the original clips and SteerVTE) are read by the same reader in the same evaluation job (Sec. 3.1), except that 99 two-row clips were re-run after a compositing fix

and their outputs re-read in a second job (re-reading 99 unchanged originals in that job reproduced 98 of their 99 verdicts). Paired differences are reported with 95% bootstrap confidence intervals over clips [8] and readability differences with exact McNemar tests [24]. Forty clips were inspected while developing the selective-editing components (Sec. 5.5); Tab. 1 reports all clips and, separately, the 486 clips that were never inspected.

Baseline. SteerVTE [42] is the strongest published video text editor; it adds glyph and style conditioning to a frozen VACE-14B backbone [16] and is trained on one million triplets. We ran the released model on every clip with the hint box as editing mask and the target as text, at the same resolution and length as the input. Its outputs are generated end to end, so they are evaluated as they are.

5.1. Main results

Table 1 compares Banksy with SteerVTE and with the unedited clips. Banksy makes 91.4% of the clips readable, 7.4 points more than SteerVTE (69 clips are correct only with Banksy, 30 only with SteerVTE; McNemar $p = 1.1 \times 10^{-4}$), and turns 211 of the 242 garbled clips into correct ones. It is not free of damage: it breaks 14 of the 284 originally correct clips where SteerVTE breaks 6, so its advantage in net repairs (197 vs. 158) is smaller than in repairs alone. The picture is unchanged on the 486 clips outside the development set (91.2% vs. 84.2%; 175 vs. 141 net repairs).

The two methods trade different qualities. Banksy leaves the background intact—its PSNR outside the text region is 6.5 dB higher (95% CI [6.2, 6.7]; higher on 98.1% of the clips)—and its strokes jitter less (edge jitter 7.85 vs. 8.34), because every glyph is placed once on the tracked carrier. SteerVTE's text region flickers less (1.66 vs. 1.79) and, above all, its text looks more like the original: its appearance distance is 1.81 against 2.50 for Banksy. We analyse this gap in Sec. 6.2.

Figure 3 breaks readability down by carrier material. Banksy is ahead of SteerVTE on six materials and level on wood; the largest gains are on signboards (+15.1 points) and screens (+10.8), where SteerVTE most often keeps the garbled characters. By language, Banksy reaches 90.5% on the 295 English and 92.6% on the 231 Chinese clips (SteerVTE: 84.1% and 84.0%). Figure 4 shows randomly sampled examples, including a failure of Banksy (花店, glass).

5.2. Edit locality

Figure 5 visualises where each method changes the video. Because SteerVTE passes the whole clip through its generator, its difference to the original covers the entire frame, including edges of buildings and furniture far from the text;

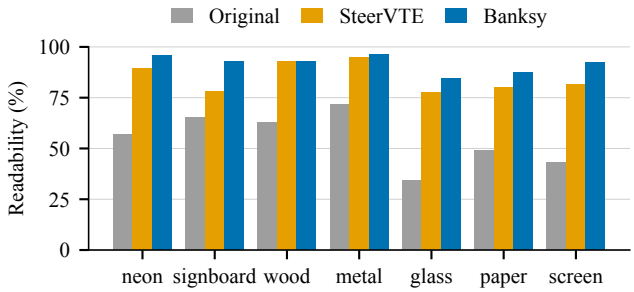


Figure 3. **Readability by carrier material** (all 526 clips; 54 to 96 clips per material).

for Banksy, the difference is confined to the text region, and the residual elsewhere (≈1 grey level) is the re-encoding of the output video. Inside the neighbourhood of the text, we additionally measure the *collateral change*—the change outside both the old and the new glyphs—which is 1.52 grey levels for Banksy; it cannot be computed for SteerVTE, whose glyph coverage is unknown.

5.3. The evidence gates

Preservation gate. The gate skips 46 clips, and the reader confirms all 46 as correct (no false skip; the pre-registered limit was 2%). It restores 2 of the 16 originally correct clips that Banksy would otherwise break, which is the entire readability difference between the two rows of Tab. 1 (2 vs. 0 discordant clips, $p = 0.5$). Its recall is low by design: it fires on 46 of the 284 clips the reader judges correct (16%), because the pipeline's OCR rarely reads stylised originals—neon italics, extruded letters, condensed faces—with the required consistency. All 14 originally correct clips that Banksy breaks are such cases (Sec. 6.1).

Carrier anchoring and style-matched rendering. Table 2 ablates the rendering components on an earlier front end. Pasting the target into the per-frame OCR box makes the text readable but nearly doubles flicker relative to the original clip (3.12 vs. 1.98), because OCR boxes jitter from frame to frame. Anchoring everything on the tracked carrier removes most of it: carrier-space erasure contributes most (-0.93), tracking -0.32 and anchored masks with ink smoothing -0.05, and the three together reduce flicker by 1.45, more than the sum of their parts, to the level of SteerVTE (1.67 vs. 1.66) while lowering jitter well below it (7.12 vs. 8.30). Readability does not suffer (89.0 vs. 87.1, $p = 0.064$). Matching the font to the old strokes adds 2.6 points of readability (D1; $p = 0.044$) once candidates are rendered in the layout of the original—an earlier version compared single-line candidates with two-line originals and systematically picked the thinnest faces. Per-frame ink and a shading field (D2) bring the appearance closest among the

Table 1. **Main results.** Readability: share of clips whose text the reader transcribes exactly in at least 4 of 5 frames. Fixed / broken: garbled → correct and correct → wrong transitions; net = fixed − broken. Appearance: glyph-agnostic material distance to the original text (normalised within this run; lower is closer). BG PSNR: outside the dilated text region. Flicker and jitter: motion-compensated, inside the text box (grey levels). Development clips (40) are excluded in the lower block. The 99 two-row clips affected by a compositing fix were re-run and re-read in a second job (supplement).

	Readability ↑	Fixed ↑	Broken ↓	Net ↑	NED sim. ↑	Appearance ↓	BG PSNR ↑	Flicker ↓	Jitter ↓
All 526 clips									
Original (generated)	54.0	–	–	–	0.819	–	–	1.98	9.18
SteerVTE [42]	84.0	164	**6**	158	0.946	**1.81**	36.49	**1.66**	8.34
Banksy w/o preservation gate	91.1	**211**	16	195	**0.962**	2.65	**42.97**	1.77	**7.73**
Banksy	**91.4**	**211**	14	**197**	**0.962**	2.50	**42.97**	1.79	7.85
486 clips outside the development set									
Original (generated)	55.1	–	–	–	0.822	–	–	1.98	9.11
SteerVTE [42]	84.2	147	**6**	141	0.945	**1.83**	36.40	**1.65**	8.29
Banksy	**91.2**	**189**	14	**175**	**0.960**	2.53	**42.93**	1.79	**7.85**

Table 2. **Rendering components** (earlier front end, 526 clips). Top: carrier-anchored temporal components; bottom: style-matched rendering. Appearance is normalised within each block. The box policy used by Banksy's fallback is T4 + D2.

	Read. ↑	Flicker ↓	Jitter ↓	Appear. ↓
Original	53.8	1.98	9.18	–
SteerVTE	83.8	1.66	8.30	–
Naive paste (per-frame OCR box)	87.1	3.12	11.66	–
+ carrier tracking (T1)	87.3	2.80	10.28	–
+ carrier-space erasure (T2)	88.6	2.19	9.27	–
+ anchored masks, ink smoothing (T3)	87.1	3.06	11.47	–
+ all three (T4)	89.0	1.67	7.12	–
SteerVTE	83.8	–	–	1.39
Naive paste	86.9	–	–	1.89
+ font matching (D1)	89.5	–	–	1.83
+ per-frame ink, shading field (D2)	89.4	–	–	1.82
+ edge blur and grain (D3)	87.6	–	–	2.01

Table 3. **Front-end variants and the localization gate** (526 clips, one evaluation job). W/L: clips correct only with the variant / only with the box policy (McNemar p). Residual: share of old ink still visible, relative to the original, on 24 held-out clips.

	Read. ↑	vs. box (W/L, p)	Jitter ↓	Appear. ↓	Resid. ↓
Original	53.8	–	9.18	–	1.00
SteerVTE	84.0	33/68, <.001	8.30	1.50	–
Naive paste	86.7	16/37, .005	12.53	2.38	–
Box policy (T4+D2)	90.7	–	7.80	2.27	0.42
Instance layout, state-machine tracking	85.2	32/61, .003	7.68	2.50	0.21
Instance layout, carrier tracking	86.9	34/54, .042	7.68	2.44	–
+ localization gate (Banksy front end)	91.1	27/25, .89	7.66	2.33	0.23

rendered variants; imitating edge blur and grain (D3) makes text both less readable and less similar, and was dropped. T4 + D2 constitutes the *box policy*, the fallback of Banksy.

The localization gate. Laying the text out on the old glyph instance should look more faithful than filling a box, and on held-out clips it does: it leaves half as much of the old ink visible (residual 0.21 vs. 0.42 of the original). On all 526 clips, however, the first instance-aligned front end was 5.5 points *less* readable than the box policy (85.2 vs. 90.7; Tab. 3). Size and colour were not the cause—forcing the box policy's colour pipeline or a minimum letter height left the gap unchanged (supplement)—but its confidence-based tracking state machine was: replacing it by carrier tracking recovers most of the gap on the clips where an instance is found (90.8 vs. 91.3), and the remaining clips are those where the instance estimate itself is unreliable. The localization gate routes exactly these clips to the box policy, which yields a front end that is as readable as the box policy (91.1 vs. 90.7; 27 vs. 25 discordant clips) while keeping the instance path's lower residual (0.23) and lowest jitter (7.66). Two re-readings of the same outputs differ by about one point (90.3 vs. 91.3 in a second job), so we claim parity with the box policy, not superiority. A subset of 100 clips had suggested a gain of one point for an earlier gated variant (95 vs. 94) that did not replicate on the full set (88.4 vs. 91.3)—a reminder that subset gates are optimistic.

5.4. Source appearance transport

Where the gate enables SAT. In the pass analysed here, the appearance gate enables SAT on 53 of the 526 clips: 23 of 93 neon clips, 11 of 65 screens, 9 of 86 signboards, 8 of 56 metal and 2 of 54 wooden signs, and none on glass or paper. Most rejections are clips without an accepted instance (203), followed by non-emissive text (133), no cross-validated gain (104) and gain in only one direction (33). Every enabled clip has a cross-validated error ratio below 0.75 in both directions, and the 15 rejected clips closest to the threshold lie between 0.756 and 0.795.



Figure 4. **Qualitative comparison on randomly sampled clips** (one English and one Chinese clip per material, drawn with a fixed seed among garbled originals outside the development set; not curated). Each cell is the same crop of frame 24; below it, the reader's transcription (✓ correct, ✗ wrong). SteerVTE produces more natural glyphs but frequently keeps or invents wrong characters; Banksy writes the target but its glyphs often look pasted (see Sec. 6.2).

Table 4. **SAT on the clips where the appearance gate enables it** (53 of 526; paired, same front-end pass). Δ: SAT − paste only, with 95% bootstrap CI.

	Paste only	+ SAT	Δ [95% CI]
Readability (%)	98.1	100	+1 clip
Appearance ↓	2.97	1.95	−1.02 [−1.70, −0.48]
Flicker ↓	2.25	2.40	+0.15, $p < 0.001$
Jitter ↓	8.62	9.42	+0.81, $p = 0.001$
BG PSNR (dB)	43.01	43.02	n.s.

Effect. On the enabled clips (Tab. 4), SAT reduces the appearance distance by a third (−1.02; better on 42 of 53 clips) for every material—screens 3.02→1.33, neon 3.03→2.02, wood 4.12→2.21—without hurting readability or the background. It costs temporal stability: flicker rises by 0.15 and jitter by 0.81, because the transported appearance follows a per-frame distance field. On the full benchmark the readability is unchanged (90.5 vs. 90.3%).

Human judgement. Because the appearance distance is only a proxy, we ran a blind study on all 53 enabled clips (Fig. 6): raters saw the original between the paste-only and the SAT version in random order and answered three separate questions. Of five raters, one gave the same side on all three questions for 98% of the clips (including the flicker question, which asks which version is *worse*) and was excluded by a rule (identical answers on at least 80% of the

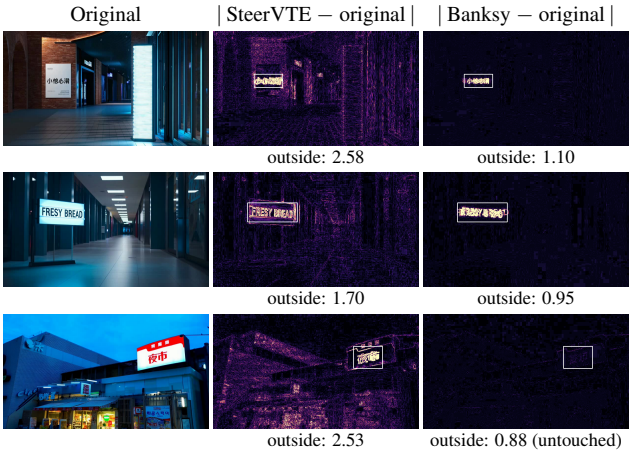


Figure 5. **Edit locality.** Absolute grey-level difference to the original frame (clipped at 24; white box: text region). SteerVTE re-renders the whole frame, so residual edges appear everywhere; Banksy changes only the text region. The numbers give the mean absolute difference outside the dilated text region. Banksy's residual there (≈1 grey level) is the video re-encoding floor, visible alone in the bottom row, where the preservation gate leaves an already-correct clip untouched while SteerVTE still alters it.

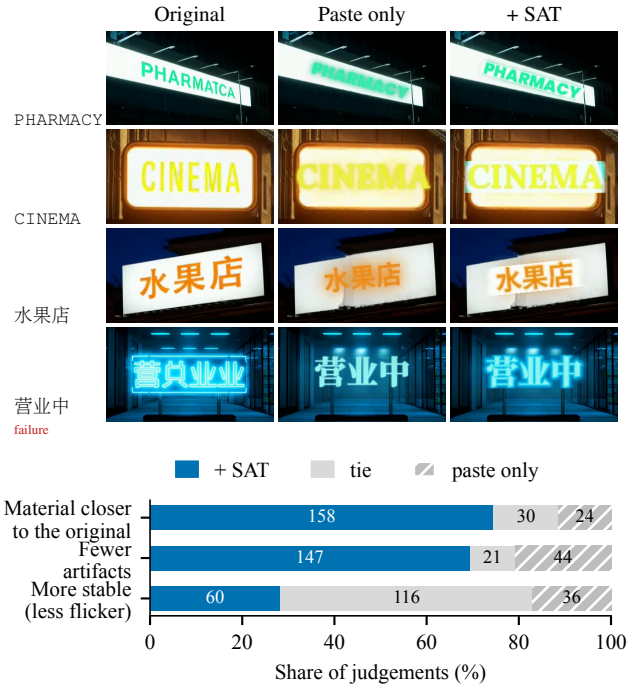


Figure 6. **Source appearance transport.** Top: SAT carries glow and material of the old glyphs to the new ones (rows 1–3: clips that raters judged better with SAT on both material and artifacts; row 4: a neon clip that all raters judged to have more artifacts with SAT). Bottom: blind pairwise study on all 53 SAT-enabled clips (4 raters, 212 judgements per question; Sec. 5.4).

clips) that we fixed only after seeing the data; we therefore also report the material question with all five raters. With four raters (212 judgements per question), SAT was judged closer to the original material in 158 cases against 24 (30 ties; win rate 0.87, 95% CI [0.77, 0.94] by bootstrap over raters and clips; 48 vs. 3 clips by majority), and as having fewer artifacts in 147 against 44 (0.77 [0.66, 0.87]); with five raters the material win rate is 0.84 [0.74, 0.92]. Agreement is moderate (Gwet's AC1 [12] 0.51 for material, 0.61 for artifacts), and the left/right position had no effect after balancing (53% left, $p = 0.27$). The measured flicker increase was *not* perceived: 55% of the flicker answers are ties, the remaining answers follow the material answer (a halo effect), and they correlate negatively with the measured flicker change ($\rho = -0.34$). The appearance distance agrees with the material votes only weakly ($\rho = 0.28$, $p = 0.049$) and not with the artifact votes, so it cannot replace human judgement.

Failures and their cause. Raters judged six clips to have clearly more artifacts with SAT, four of them neon (Fig. 6, last row): a bright rectangular plate behind the new letters. A first hypothesis—that the new glyphs reach distances never observed on the old ones—was rejected (the share of unobserved distances is at most 2.3%), and tapering the operator to identity at the edge of its support had no effect. The plates coincide instead with the canvas on which the new glyphs' distance field is computed: the canvas leaves only 0.04–0.21 letter heights of margin, so the halo, which should extend to 0.6, is cut at the canvas border. With an untruncated distance field the straight edges disappear in our diagnosis, and computing the distance field on a padded canvas reduces flicker by 0.08 and jitter by 0.37 (both better on 47 of 53 clips, $p < 0.001$) with no significant change in appearance—about half of SAT's flicker cost and 40% of its jitter cost. Because these 53 clips were used to diagnose the failure, the padded version is not part of the evaluated Banksy and requires validation on unseen clips.

5.5. Selective, structure-preserving editing

Inspecting the outputs of the first full evaluation, we found that the most frequent problem of Banksy was not that its glyphs lacked material but that it changed things it should not have: already-correct text, sign frames, and the structure around the letters. We therefore tested three components that make editing more selective, each added to B0 (Banksy without the preservation gate) on its own, with acceptance criteria fixed before the run (supplement): the preservation gate of Sec. 4.2 (B1), *glyph-only erasure* (B2), and a *carrier-fit check* (B3).

Glyph-only erasure (B2). Instead of inpainting the whole letter band, B2 erases only the old glyphs and their own

Table 5. **Selective, structure-preserving editing** (pre-registered; 526 clips, paired front-end pass; the 99 two-row clips re-run after a compositing fix). B0 is Banksy without the preservation gate; each component is added to B0 separately and then all three together (B4). Collateral: change outside the old and new glyphs in the text neighbourhood. The verdict applies the pre-registered criteria; development clips are excluded in the right block.

	All 526 clips									486 clips (no development clips)		
	Read. ↑	Fixed	Broken	Net	Appear. ↓	BG PSNR	Collateral ↓	Flicker	Verdict	Read. ↑	Net	Collateral Δ vs. B0
B0: Banksy w/o preservation gate	91.1	211	16	195	2.65	42.97	1.55	1.77	–	90.9	174	–
B1: + preservation gate (= Banksy)	**91.4**	211	**14**	**197**	2.50	42.97	1.52	1.79	adopted	**91.2**	**175**	−0.031 [−0.06, −0.01]
B2: + glyph-only erasure	83.8	180	23	157	2.53	43.06	1.46	1.80	rejected	84.6	143	−0.036 [−0.09, +0.03]
B3: + carrier-fit check	89.0	198	**14**	184	2.52	42.95	1.54	1.78	rejected	89.1	165	−0.004 [−0.01, −0.00]
B4: all three	83.1	171	18	153	**2.30**	43.05	**1.43**	1.83	rejected	83.5	138	−0.062 [−0.12, +0.00]

glow, estimated with a mixture model of carrier, ink and halo colours, and protects sign frames, long bars and the boxes of other text. It is active on the 325 clips with an accepted instance—in 117 of them it falls back to band erasure minus the protected structures, because the surroundings themselves look like strokes (*e.g.* neon stripes)—and its erased area is a median 0.64 of the band area. *Pre-registered criterion*: collateral change decreases ($p < 0.05$) and readability drops by at most one point.

Carrier-fit check (B3). B3 estimates the extent of the carrier from colour, lightness and texture boundaries and declares a clip *layout-infeasible* if the target, at the character density of the source, would have to shrink below 0.7 of the source height to fit; such clips are left unedited and counted as failures. *Criterion*: the infeasible flags are confirmed by manual inspection.

Results. Table 5 reports all variants from one front-end pass (with the 99 two-row clips re-run after a compositing fix that affects all variants alike). The preservation gate meets its criteria (no false skip; two fewer broken clips; readability not lower) and is adopted, which makes B1 the Banksy of Tab. 1. Glyph-only erasure reduces collateral change on all clips (-0.090, 95% CI $[-0.17, -0.02]$, $p = 0.017$) but not significantly once the development clips are removed ($p = 0.23$), and it costs 7.3 points of readability: the instance ink mask is not a complete glyph mask, so parts of the old strokes survive next to the new ones, and OCR boxes of the garbled text itself were sometimes protected as "other text". A second version that completes the old glyphs by colour still left remnants in three of nine diagnosed clips, which suggests that this component needs a learned stroke segmentation. Of the 24 clips flagged infeasible by the carrier-fit check, manual inspection confirmed 5, was undecided on 3 and rejected 16 (OCR missed letters of the source, or two-row signs were treated as one row), so B3 is rejected as well; a stricter version that only flags clips whose source stably shows a whole-word part of the target behaved correctly on the diagnosed clips but has not been run at scale. All three together (B4) inherit B2's readability loss. A blind study comparing B1 with SteerVTE (60 clips) and B2 with B0 (28 clips) on naturalness and on damage to non-target content is in progress (results pending).

Table 6. **Does a learned refiner help?** The 1.3B bridge-flow refiner applied on top of Banksy's front end (paste only) and of the box policy, with and without the carrier-anchored attention bias ($\lambda = 2$). 526 clips, one evaluation job; W/L against the same front end without the refiner. Boundary jump: change at the refiner's 17-frame chunk seams relative to other frame pairs (1 = no seam).

	Read. ↑	W/L	Flicker ↓	Jitter ↓	B. jump ↓	Appear. ↓
SteerVTE	84.2	–	1.66	8.34	0.93	1.58
Banksy front end (paste only)	91.3	–	1.77	7.69	0.84	2.49
+ refiner	87.8	3/21	1.96	8.22	2.26	2.45
+ refiner + attention bias	88.6	3/17	2.00	8.41	2.06	2.33
Box policy (paste only)	91.4	–	1.77	7.83	0.84	2.35
+ refiner	87.6	2/22	1.99	8.46	2.31	2.31
+ refiner + attention bias	88.4	3/19	2.03	8.68	2.10	2.24

5.6. Does a learned refiner help?

Rendered glyphs look pasted, so the obvious remedy is to let a generative model harmonise them with the scene while keeping their shapes. We invested heavily in this route before settling on the training-free pipeline, and report it in full because its outcome shapes our design.

Refiner. The refiner is a video diffusion transformer initialised from Wan2.1-T2V-1.3B [31] that takes the pasted clip as extra input channels (zero-initialised) and is trained as a flow-matching bridge [18, 20] from the paste to the target; inference starts from the paste, in the spirit of SDEdit [26], and takes 30 Euler steps on 17-frame chunks of 384×384 crops. A glyph loss compares Sobel edges of the decoded prediction with the target inside a band around the glyph (weighted by the signed distance to the glyph boundary). Training pairs come from three sources: a Blender data factory with six carrier surfaces and passing occluders, which renders every sample twice (lit composite and flat paste); 562 real pairs obtained by erasing correct text from generated clips and re-pasting it; and up to 1.15 million image pairs built the same way from AnyWord-3M (63% LAION, 31% Wukong, 6% street scenes). All refin-

ers were trained on H100-class GPUs (GH200/H200) for 20k steps.

More data is not the lever. On a 296-clip garbled subset, refiners trained with 0, 10k, 100k and 1M image pairs reach 78.4, 75.3, 76.4 and 76.4% readability, all below the paste they start from (80.7%), and their appearance does not improve with data (supplement). A diagnostic on 1,500 held-out pairs explains why: on the very task they were trained for—turning the paste back into the original—all of them move *away* from the original, they do not even memorise pairs seen twelve times, and they fail to learn a systematic difference (originals have thicker strokes than our pastes in 60% of the pairs). The supervision was inconsistent: the paste uses our font, the target uses the generator's, and the glyph loss, which dominates the objective, effectively teaches the model to erase our strokes. Making the supervision glyph-consistent—pasting the original's own ink mask instead of our font—is the first recipe that moves toward the original on held-out pairs and it cuts the image-domain readability loss from 26 to 4 points, with appearance better than the paste. On videos, however, it still does not exceed the paste (best variant: 79.7 vs. 80.7%).

System-level effect. Table 6 applies the best refiner to the full benchmark. On top of Banksy's front end it costs 3.5 points of readability (21 clips broken against 3 repaired; $p < 0.001$), increases flicker and jitter, and introduces seams at its chunk boundaries (boundary jump 2.26 vs. 0.84), while the appearance distance does not change significantly (-0.035, 95% CI $[-0.21, 0.19]$). A training-free attention bias that ties tokens to their carrier trajectory, in the spirit of flow-guided attention [6, 10], was selected on 50 clips with an earlier refiner (boundary jump -16% there) and reduces the seams by 9% on the full set (2.06 vs. 2.26; better on 364 of 526 clips), but it slightly increases flicker within chunks and leaves the readability loss in place. We therefore keep the refiner out of Banksy. The negative result is specific to our scale—1.3B parameters and at most a few hundred real video pairs, against a 14B backbone and a million training triplets for SteerVTE—but it is consistent: at this scale, a learned harmoniser trades the guarantee that motivates rendering for an appearance gain we could not measure.

5.7. External benchmarks

Both external evaluations below were run with the earlier box-policy front end (Tab. 2); re-running them with the final pipeline is pending, and we report them as they are.

Editing correct text (VTE-Bench). On the 100 real clips of SteerVTE's own benchmark [42], where correct text must be changed into a different string, SteerVTE reaches 92% readability and our earlier pipeline 82% (8 vs. 18 discordant clips, $p = 0.076$; a second reader, EasyOCR, gives 49% vs. 44%). Editing differs from repair: the source text is valid and must be removed entirely, which is the regime that generative editors are trained for and in which our gates provide no advantage. We include the result to delimit our claims.

Commercial video models (VidScribe). VidScribe [46] releases the outputs of seven commercial video models for video-to-video text editing. On the 44 of its 93 clips that have a localisable text region, our earlier pipeline reaches 29.5% (naive paste) and 27.3% (box policy) readability, ahead of the strongest commercial model (minimax-h3, 18.2%) and wan-3.0 (15.9%), although not significantly (box policy vs. minimax-h3: 6 vs. 2 discordant clips, $p = 0.29$); against the other five models (0–13.6%) the box policy is significantly better ($p \leq 0.031$). Commercial models regenerate the whole clip and lower the PSNR outside the text region to 19.8–27.9 dB, against 37.6–37.9 dB for our outputs. Absolute readability is low for every method because most VidScribe targets are long phrases.

6. Analysis and limitations

6.1. Where Banksy fails

Breaking correct text. Banksy breaks 14 of the 284 originally correct clips (SteerVTE: 6). In all of them, the evaluation reader transcribes the original perfectly while the pipeline's OCR reads it correctly in almost no frame (neon italics, extruded or condensed letters), so the preservation gate cannot fire. Inspecting the outputs (Fig. 7, last three rows) gives five failure types: new text larger than its sign, so that letters fall off the carrier (3 clips); wrong colour or contrast on lightboxes and extruded letters (2); a second row that was not detected, leaving the old text visible next to the new one (4); a neon glyph rendered as a blob (1); and stacked two-row text that is still misread (2). The last two had their first row partly erased by a compositing error in the box policy; after the fix, both rows are rendered completely, but one is still read as `SALE -30%` on an LED screen and the other is misread in two of five frames. The remaining 2 clips are borderline cases in which the reader additionally transcribes small neighbouring text in 2 of the 5 frames. Across all 99 two-row clips the fix changed the verdict of only two clips (one repaired, one lost), so the damaged glyphs mostly co-occurred with other failures.

Readable but not faithful. The more common failure does not affect readability (Fig. 7, first three rows): erasure leaves a visible patch, or the new glyphs are flat where the old ones were outlined, extruded or glowing. These cases



Figure 7. **Failure modes of Banksy** (hand-picked to cover each type; SteerVTE shown for reference). The first three keep the target readable but degrade appearance; the last three come from clips whose original was already correct and that Banksy broke.

are frequent in the random sample of Fig. 4 and explain the appearance distance in Tab. 1.

6.2. The appearance gap to generative editing

Among the 148 garbled clips that both methods repair, SteerVTE's text is closer to the original appearance in 70% (median appearance distance 0.92 vs. 1.72), so the gap is not an artefact of SteerVTE leaving garbled text untouched. We see four causes, in decreasing order of visibility.

- **Rendering instead of generation.** Banksy can reproduce colour, shading and, with SAT, glow; it cannot reproduce extrusion, bevels, tube geometry or texture inside the strokes, which a generator synthesises naturally.
- **A small font library.** The library holds 33 Latin and 8 Chinese families, and the generator's typefaces are invented. In practice, 26 Latin families are selected, but for Chinese only three, and Noto Sans SC alone renders 68% of the Chinese clips; the result reads correctly but looks uniform.
- **Axis-aligned layout.** The letter band inherits the orientation of the OCR quad, which is axis-aligned for most detectors, so on tilted carriers the new text can look skewed relative to the sign (Fig. 1a); the perspective estimate of an earlier front end was too fragile to keep (supplement).
- **Erasure traces.** LaMa fills the hole once per clip; on textured or shaded carriers the fill is visible as a patch.

Two observations suggest that the gap is closable without giving up correctness. First, the garbled source already contains most of the target's characters: in garbled English clips, a median of 88% of the target letters occur in the reader's transcription of the original (Chinese: 67%). Reusing these glyphs—cut from the source with their own font, colour, lighting and perspective—and synthesising only the missing ones would extend the "change only what is wrong" principle from words to characters. Second, the two methods fail on different clips: the share of clips that at least one of them repairs is 97.1%. A combination that accepts a generative edit only when independent OCR verifies it, and otherwise falls back to deterministic repair, is a natural next step; its verifier must be strict, since accepting a wrong edit costs correctness, whereas rejecting a right one costs only appearance.

6.3. Other limitations

The benchmark is generated by one family of models and its prompts ask for short sign texts; a second generator (CogVideoX-5B, 360 clips from the same prompts) is part of our plan but has not been evaluated yet, and long texts such as menus or user interfaces are out of scope. Readability is measured by one VLM reader; it favours large, high-contrast text, which may flatter the box policy, and a second reader is needed. The human study on SAT used four raters after quality control, and the study comparing Banksy with SteerVTE on naturalness and on damage to non-target content is still being collected. Finally, the preservation gate is safe but has low recall, because it relies on a conventional OCR that rarely reads stylised originals.

7. Conclusion

Repairing garbled text in generated videos is not the same problem as editing text: the source is a near miss, it is often already right, and everything around it should survive. Banksy treats it as a sequence of evidence-gated decisions—leave the clip alone, lay the text out on the old glyphs or on the tracked box, transport the old appearance or not—around a rendering step whose spelling cannot drift. This makes it more reliable than the strongest generative editor on our benchmark and confines its changes to the text, but its glyphs still look rendered rather than generated. Our analysis points to two ways of closing that gap without giving up correctness: reusing the characters that the generator already drew correctly, and accepting generative edits only when they can be verified.

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Banksy: Evidence-Gated Repair of Garbled Text in AI-Generated Videos

Supplementary Material

A. Benchmark details

Generation. Clips were generated with the text-to-video mode of VACE-14B from 120 prompts, each asking for a scene with a sign of a given material that reads a quoted target string, with ten seeds per prompt; 1,178 of the 1,200 generations were usable (832×464, 49 frames, 15 fps). An OCR screen confirmed the requested text in 24% of them (English 21%, Chinese 27%). The 526 benchmark clips were selected as garbled by an EasyOCR-based screen (selection rule to be documented); our evaluation reader judges 54.0% of them correct, so the benchmark contains repair and do-nothing cases.

Targets and composition. The 40 target strings are: `BUS STOP` (25 clips), `FRESH BREAD` (23), `BOOK STORE` (22), `PHARMACY` (22), `COFFEE SHOP` (21), `TEA HOUSE` (20), `NO PARKING` (19), `FLOWER MARKET` (19), `SALE 50%` (18), `NIGHT MARKET` (18), 面包房 (17), 小心地滑 (17), 邮局 (16), `ICE CREAM` (15), 停车场 (15), 电影院 (15), 书店 (14), `RECORDS` (13), 出口 (13), 洗衣店 (13), 咖啡店 (12), 茶馆 (12), 欢迎光临 (11), 营业中 (11), `OPEN` (10), `EXIT` (10), 理发店 (10), `GALLERY` (9), 花店 (9), 公园 (9), 火锅 (9), `BAKERY` (8), 夜市 (8), `HOTEL` (7), 便利店 (7), 药房 (7), `LAUNDRY` (6), 水果店 (6), `CINEMA` (5) and `WELCOME` (5). Table 7 gives the material × language composition and the readability of each method per group.

Table 7. Composition of the benchmark and readability (%) per group, final run.

Group	EN	ZH	Original	SteerVTE	Banksy
neon	47	46	57.0	89.2	95.7
signboard	74	12	65.1	77.9	93.0
wood	20	34	63.0	92.6	92.6
metal	22	34	71.4	94.6	96.4
glass	52	24	34.2	77.6	84.2
paper	55	41	49.0	80.2	87.5
screen	25	40	43.1	81.5	92.3
English (295)			62.4	84.1	90.5
Chinese (231)			43.3	84.0	92.6

Development data. We disclose every clip that was inspected during development. (i) Twelve clips ("dev12") were used to tune the instance front end and the SAT oracle; 24 random clips ("held24") were not used for tuning, but screenshots of four of them were viewed during diagnosis and held24 informed the SAT margin, so they are not strictly held out. (ii) The 53 SAT-enabled clips were used for the SAT human study and then for the failure diagnosis of Sec. 5.4; results of the padded canvas on them are development results. (iii) Forty clips were inspected while developing the selective-editing components; these are the clips excluded in the right block of Tabs. 1 and 5: `g001_en_neon_s1 g006_en_paper_s3 g007_en_glass_s6 g008_en_signboard_s0 g010_en_neon_s9 g011_en_wood_s1 g020_en_glass_s5 g020_en_glass_s7 g022_en_neon_s1 g024_en_metal_s0 g025_en_paper_s4 g028_en_signboard_s3 g029_en_screen_s2 g036_en_neon_s6 g045_en_signboard_s8 g046_en_neon_s2 g046_en_neon_s4 g046_en_neon_s5 g049_en_wood_s5 g052_en_screen_s4 g057_en_neon_s6 g057_en_neon_s7 g059_en_metal_s2 g067_zh_wood_s9 g071_zh_neon_s0 g076_zh_neon_s5 g076_zh_neon_s8 g083_zh_neon_s1 g084_zh_neon_s0 g084_zh_neon_s3 g093_zh_neon_s7 g098_zh_neon_s1 g107_zh_neon_s2 g107_zh_neon_s9 g110_zh_neon_s0 g110_zh_neon_s4 g110_zh_neon_s5 g110_zh_neon_s7 g110_zh_neon_s8 g116_zh_screen_s8.` The main conclusions do not change when they are removed.

B. Evaluation details

Reader. Qwen2.5-VL-7B transcribes the hint-box crop (15% padding) of frames 0, 12, 24, 36 and 48. A frame is exact if its normalised edit similarity to the target is at least 0.999 after normalisation (whitespace removed, lowercased, traditional Chinese folded to simplified); a clip is correct if at least 4 of 5 frames are exact. An earlier draft of the pre-registration described this as "at least 60% of frames"; this was a documentation error, the code was fixed before the run, and the decisions are unaffected.

Evaluation hygiene. In an earlier phase, the reader produced garbled transcriptions (a repeated token) for about 8.5% of the inputs on one cluster (x86, an older version of the inference library) and none on another (aarch64, newer version). Two refiners had been read on the clean system and their baselines on the faulty one, which inflated a comparison by 16.6 points; after re-reading everything in one job the advantage disappeared (Appendix E). Since then, every table is produced by re-reading all of its rows in a single job, and we check transcriptions for garbage. Re-reading identical outputs in a second job changes readability by about one point, which is why we treat differences of that size as ties: when the 99 unchanged originals of the two-row re-run were read again, 75 received identical transcriptions and 98 the same verdict. The front end it-

self is not bit-reproducible across runs (GPU OCR boxes, LaMa on different devices); comparing two passes of identical code gives no significant difference in appearance or flicker but a 0.46 dB difference in background PSNR, so we only compare variants from the same pass.

Appearance distance. Features per frame (CIELAB, inside the box): ink colour difference to the original, ink–ground contrast difference, halo excess normalised by contrast, boundary gradient normalised by contrast (edge softness), and standard deviation of lightness inside the eroded strokes (inner shading). The original compared with itself scores about 0.27, the estimation floor. Each feature is divided by its median over all method–clip pairs of the run, so values are comparable only within one evaluation job.

C. Implementation details

Reference frame. Among the frames whose OCR score is at least 90% of the maximum, the one closest to the middle of the clip.

Gates. Preservation gate: complete target read in $\geq$85% of the frames, wrong readings in $\leq$5%, reading inside the hint box, extra boxes in $\leq$10% of the frames, and the concatenated reading of all boxes inside the hint box equal to the target in $\geq$90% of the correct frames. Localization gate: $\geq$3 letters ($\geq$2 for Chinese), CIELAB distance between ink inside the letter band and inter-letter ground $\geq$40, letter height between 0.3 and 1.0 of the OCR box height, no fallback row grouping; otherwise the box policy is used (201 of 526 clips in the final run; in an earlier pass, the reasons were the row grouping for 134, letter height for 67, letter count for 40 and contrast for 36 clips, with overlaps). Appearance gate: emissive text (ink brighter than ground, or glow detected on the carrier) and a cross-validated error ratio below 0.75 against the flat fill in both directions.

Letter height and polarity. Latin letter height is the median height of letter-like components (solidity $\geq$0.85); Chinese letter height is taken from the ink distribution. Ink and ground are separated by k-means with $k \in \{2, 3\}$ scored by letter morphology (component heights and holes), with a fixed seed.

Font library. 33 Latin and 8 Chinese families from Google Fonts, variable fonts expanded over their weights. In the final run, 26 Latin families and 3 Chinese families were selected (Noto Sans SC for 68% of the Chinese clips, Noto Serif SC for 21%, ZCOOL XiaoWei for 12%).

SAT. The operator of Eq. (1) is fitted per distance bin (signed distance in letter heights, support up to 0.6 outside the strokes) by least squares on the reference frame and applied to every frame through the carrier homography. The flat fill is the ink colour of the box policy. Cross-validation fits on the left half of the text row and tests on the right half, and vice versa.

Runtime. On an NVIDIA A40, the earlier front end took about 20 s per clip and the learned refiner about 40 s per clip; SteerVTE reports 108 s per 49-frame clip on its hardware. Runtime of the final pipeline to be measured.

D. Pre-registration of the selective-editing study

The study was registered before the run in an internal document with dated revisions. **Criteria.** B1: fewer broken clips, false skips $\leq$2% of the skipped clips, paired readability not lower than B0 by more than 0.5 points. B2: collateral change decreases ($p < 0.05$), readability not lower than B0 by more than one point, and the human artifact question favours B2. B3: infeasible flags confirmed by manual inspection; results reported with infeasible clips counted as failures (main) and excluded (reference). Components are attributed separately before B4 is considered. **Revisions during development** (on the development clips only, before the run) concerned the B2 glow model and the B3 boundary cues. **Revisions during the run** (before any result was seen, recorded at the time): B3 feasibility was changed to use the character density of the source, measured on the best reading over all frames; B1 received its fifth condition after two false skips were found while checking decisions (a correct word above garbled small text; a near-miss reading split across boxes). All 526 clips were then re-run with the final code. **Round two** (defined after the results, before any human data): a colour-completion variant of B2 and a whole-word-subset variant of B3 were tested on nine and six diagnosis clips, respectively, and were not run at scale. **Excluding infeasible clips** (reference): readability B1 92.0, B3 91.8; after manual review, replacing the non-confirmed B3 flags by B0 outputs gives 90.3%.

E. Additional results

Diagnosing the instance front end. On the 379 clips where an earlier gated variant accepted an instance, forcing the box policy's colour pipeline (88.4%) or a letter height of at least 0.6 box heights (88.1%) did not close the gap to the box policy (92.3%); replacing the box erase did not help either (87.6%), whereas replacing the tracking state machine by carrier tracking did (90.8 vs. 91.3%). A refinement with row splitting from the ink profile, a perspective trapezoid fitted to letter tops and bottoms (only with $\geq$6 homogeneous letters and consistent fits) and shadow-aware erasure fixed

several stacked and perspective cases visually but lowered readability on aligned clips by 1.6 points and was not kept.

Material-conditioned rendering. We also tried rendering models per material (glow for neon and screens, bevel and shadow for metal, paint following the texture for wood, paper and signboards, translucency for glass), selected by the prompt label or by a VLM prediction. A wrong material model hurts consistently (appearance $+0.28$, $p < 0.001$; worse on six of seven materials), but the right one helps clearly only on screens (-0.49, $p < 0.001$) and hurts on glass ($+0.30$). The prompt label agrees with the visual material in only 60.5% of the clips (the generator often renders signboards and screens as glowing text). An automatic pairwise realism judge (the same VLM, both orders averaged) scored 0.49–0.51 for every pair and could not discriminate. These models were dropped; SAT replaces them for emissive carriers.

Occluders. Carrier anchoring assumes that nothing passes in front of the text. On 90 garbled clips with synthetic occluders of known alpha (30 each crossing, fully covering for 13 frames, or covering half of the text), the temporally anchored pipeline erased or over-painted occluders more than naive pasting (occluder damage 10.3 vs. 5.4 grey levels). An occlusion map from a readable-frame median background with per-pixel adaptive thresholds reduced the damage to 3.4 (-66%) but also lowered readability on visible frames (67% vs. 73%) because of false positives on strokes. Occlusion handling is therefore not part of Banksy, and clips with occluders remain a limitation.

Learned refiner. Table 8 summarises the refiner study on the 296-clip garbled subset (window-level readability, all rows read in one job) and on 1,500 held-out image pairs. "Closed" is $1 - \sum |\hat{I} - I| / \sum |P - I|$ over the text region, where P is the paste and I the original (0: equal to the paste; 1: perfect).

Combining with a generative editor. An earlier version of our system used SteerVTE as an editor inside a verify-and-select loop: skip clips whose text is already correct, try SteerVTE first, accept its output only if EasyOCR reads the target, otherwise try our earlier refined rendering, and finally a second SteerVTE seed. On the 526 clips it reached 91.1% readability, 82.2% on the garbled clips and kept 98.6% of the correct clips correct, against 84.0%, 67.8% and 97.9% for a single SteerVTE run (earlier front end; the reader-selected best of two SteerVTE seeds reaches 89.4%). This supports the complementarity noted in Sec. 6.2; its appearance was not evaluated.

Table 8. Learned refiner: data scale and supervision. Video: 296 garbled clips (Lab dist.: mean CIELAB distance to the original in the original ink region; lower is closer); image: 1,500 held-out pairs (our-font paste; "closed†" with the original's own glyph pasted).

	Video		Held-out image pairs		
	Read.	Lab dist.	Read.	Closed	Closed†
Paste	80.7	53.9	81.3	0	0
Refiner, 0 image pairs	78.4	51.1	68.9	−0.121	–
10k pairs	75.3	51.7	51.9	−0.059	–
100k pairs	76.4	51.8	55.5	−0.064	−0.491
1M pairs	76.4	53.1	52.7	−0.058	–
Glyph-consistent, 100k	77.7	52.8	77.0	−0.026	+0.049
Glyph-consistent, 1M	78.7	52.7	76.5	−0.029	–
+ video pairs, street ×4	79.7	52.0	76.7	−0.029	–

Neon structure reconstruction. On the 23 SAT-enabled neon clips we explored neon-specific operators—keeping the sign frame out of erasure and SAT, rebuilding double-line tube geometry, and restricting the rendered text to the word visible in the source. A non-blind preview judged three of six examples good and three faulty (old content showing through, a changed font style, and invented text); after fixes, this line was paused in favour of selective editing.

F. Human studies

SAT study. All 53 SAT-enabled clips; each trial shows the paste-only and the SAT version left and right of the original (full frame plus an enlarged text row), with the side randomised per clip (SAT on the left in 26 of 53). Raters, who know the target, answer separately: which version's material is closer to the original, which has fewer artifacts, and which flickers more (ties allowed). Five raters completed the study; the exclusion rule and its sensitivity analysis are described in Sec. 5.4. Analyses use win rates with ties excluded, two-way bootstrap over raters and clips, clip-level sign tests, Fleiss' κ and Gwet's AC1.

Selective-editing study (in progress). 88 pairs: Banksy (B1) vs. SteerVTE on 60 clips sampled with a fixed seed among the 480 clips that Banksy edits, stratified by material × outcome (both correct, only one correct, neither) with proportional allocation (1–4 per cell), and B2 vs. B0 on 28 clips sampled among the 285 clips where B2 changes the erasure. Questions: which version's text belongs more naturally to the original scene, and which version damages the non-target content less; correctness is recorded separately. Left/right assignments are stored separately from the study page.

G. Failure cases

The 14 originally correct clips broken by Banksy, by type: stacked two-row text still misread after the compositing fix (`SALE 50%` on an LED screen, `BUS STOP`); text larger than its carrier (`BOOK STORE` on neon, `GALLERY`, `HOTEL`); wrong colour or contrast (书店 on extruded letters, `WELCOME` on a lightbox); undetected second row or remaining old text (`TEA HOUSE`, `FLOWER MARKET` ×2, 面包房); blurred neon glyph (出口); and two borderline reader cases (`ICE CREAM`, 火锅) in which small neighbouring text is transcribed in 2 of 5 frames.